EXPERIMENTAL INVESTIGATIONS OF GRID FIN AERODYNAMICS: A SYNOPSIS OF NINE WIND TUNNEL AND THREE FLIGHT TESTS

Wm. David Washington
U.S. Army Aviation and Missile Command
Attn: AMSAM-RD-SS-AT
Redstone Arsenal, Alabama 35898-5000, USA

Mark S. Miller
Dynetics, Inc., P.O. Box 5500
Huntsville, Alabama 35814-5500, USA

SUMMARY
Wind tunnel and flight tests have been conducted to investigate the aerodynamic characteristics of grid fins and to demonstrate their utility in actual flight hardware and ballistic flight performance. Nine wind tunnel tests have been conducted on 26 different grid fin configurations. Test parameters have included Mach numbers ranging from 0.35 to 3.5 with nominal angles of attack up to 17 degrees. Investigative issues have included: basic aerodynamic coefficients, grid fin curvature for efficient packaging, drag reduction techniques, transonic choke regions, geometric variables (span, chord, height, cell spacing and web thickness), fin sweep back effects, and grid fin/planar fin comparisons.

Two flight tests, with rockets launched out of a circular launch tube, successfully demonstrated the capability to package and stow grid fins within a circular rocket body shape, deploy grid fins during flight, despin, and stabilize the warhead section after stage separation until completion of the mission. An additional air drop flight test was conducted to demonstrate the flight worthiness of a rocket payload section during aft dispense of multiple submunitions.

1. INTRODUCTION

The Research Development and Engineering Center of the U. S. Army Aviation and Missile Command, Huntsville, Alabama has investigated the aerodynamics of grid fins since 1985. A total of nine wind tunnel tests have been conducted in order to gain greater insight into the aerodynamic characteristics of grid fins for Mach numbers ranging from 0.3 to 3.5. Table 1 summarizes the objectives of each test. As indicated in Table 1, six wind tunnel tests were dedicated to understanding the unique aerodynamic characteristics of grid fins. An additional two tests were focused on evaluating specific point designs as part of the Bat-On-A-Rocket (BOAR) missile technology demonstration program.

The grid fin is a unique device that can be used as either an aerodynamic stabilizer or a control surface. Its unique design and aerodynamic characteristics separate it from conventional planar fins. Planar fins can generally be described with information about the root chord, tip chord, span and thickness. Grid fins, however, possess an extra dimension. Grid fins require five geometric parameters to describe their geometry (span, chord, height, cell spacing and element thickness). An attempt has been made through a series of tests to understand the importance of each of these parameters. A description of test articles and a synopsis of findings are presented in this extended abstract.

2. TEST ARTICLE DESCRIPTION

A total of 26 separate grid fin designs have been evaluated to identify critical design parameters. Table 2 provides a listing and configuration code for each of the designs tested; an accompanying photograph of these designs is presented in Figure 1. Many of the designs listed in Table 2 have been tested while mounted on a 10.4 caliber body of revolution with a three caliber tangent ogive nose section, followed by a 7.4 caliber cylindrical section. (See Figure 2). The 5-inch diameter body was mounted on a six component main balance for all tests. Four fin balances were attached to the aft section, 1.2 calibers forward of the model base. Both five-component (measuring all components except fin drag) and six-component fin balances have been utilized. Test conditions generally ranged from Mach 0.5 to 2.5, with a limited amount of data taken at Mach 3.5. Additional

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subsonic data have also been obtained on grid fins mounted on a splitter plate to measure isolated six-component fin data. Unless otherwise noted, all grid fin aerodynamic coefficients presented are based on fin balance measurements (obtained in the presence of a body) with a consistent reference area of 19.635 in^2 (body cross sectional area).

3. SYNOPSIS OF TEST RESULTS

3.1 High Angle of Attack Characteristics

One of the advantages of grid fins is their capability to produce effective aerodynamic control forces at high angles of attack over wide Mach number ranges. Unlike conventional planar fins, grid fins do not experience classical "stall" at higher angles of attack. This characteristic leads to more effective stability and control characteristics at intermediate and large angles of attack. Figure 3 presents splitter plate (fin alone) data obtained on two grid fin designs at Mach 0.35. Maximum normal force occurs at approximately a 40 degree angle of attack. There is no characteristic stall break which is typical of planar fins at moderate angles of attack.

Figure 4 presents grid fin control characteristics (fin balance data obtained in the presence of a body) at subsonic, low supersonic and high supersonic speeds. Note the almost linear characteristic with angle of attack for supersonic Mach numbers. This characteristic indicates that grid fins are very effective control devices for missiles at supersonic speeds. A comparison with planar fins (Section 3.8) shows grid fins to be more effective than planar fins at supersonic speeds.

3.2 Hinge Moment / Packaging Advantages

Two other advantages of grid fins relative to planar fins are: small hinge moments and excellent packaging characteristics. Figure 5 presents hinge moment coefficients as a function of angle of attack for flat, concave and convex grid fins tested at subsonic, transonic and supersonic Mach numbers. Curved grid fins were fabricated with a radius of curvature equal to the 2.5-inch radius of the cylindrical section of the wind tunnel model shown in Figure 2. The projected planform of each grid fin was exactly the same. Results indicate that hinge moments are very small. The small hinge moments generated by grid fins are very attractive for missile applications. Smaller actuators are less expensive and require less control power and internal volume.

Additional findings are that curved grid fins (both concave and convex) and flat grid fins produce essentially the same aerodynamic forces. Variations in normal force due to curvature are barely distinguishable. These results give greater flexibility for packaging in practical missile design. Curved grid fins can be easily stowed within the moldline of a missile body by folding forward or backward and then deployed after launch or staging for multi-stage missiles. This concept was demonstrated in flight tests discussed later in the paper.

3.3 Drag Reduction Techniques

One of the most serious concerns related to the utilization of grid fins is the likelihood of increased drag over an equivalent planar fin. Six grid fins were designed and tested to examine the drag characteristics of grid fins and evaluate various techniques for reducing drag levels. One of the easiest approaches for reducing drag levels is to alter the cross section shape of the outer frame design. Figure 6 illustrates the different frame cross sections tested. Geometric parameters were frame shape, web thickness and frame thickness. Note that the grid fin planform shape was unchanged for these models. Results from the test are shown in Figure 7 for Mach numbers of 0.7 and 2.5. At least a 25% reduction in fin drag is realized by altering the shape of the grid fin frame.

3.4 Transonic Aerodynamics

Figure 8 presents a plot of grin fin normal force coefficient slope versus Mach number. Unlike conventional planar fins which experience maximum normal force coefficient values at transonic Mach numbers, grid fins experience what is commonly referred to as a "transonic bucket". To understand this phenomenon, one must look at a grid fin as a collection of individual cells acting as separate inlets.

Referring to Figure 9, the reduction in the inlet cross-sectional area, created by the presence of the cell walls and the boundary layer buildup on the walls, causes the flow passing through the cell to accelerate to sonic conditions (i.e., the flow becomes choked) at freestream Mach numbers less than 1.0. As the flow increases beyond Mach 1.0, a detached normal shock forms in front of the cell. The cell remains choked until the normal shock is "swallowed". It is the flow spillage from the choked flow that causes the reduction in normal force. At higher Mach numbers, a leading edge shock passes through the cell undisturbed and the grid fin exhibits normal force characteristics similar to conventional planar fins.

3.5 Design Parameter Variation

One of the six research wind tunnel tests examined the influence of grid fin geometric parameters for Mach numbers ranging from 0.7 to 2.5. Five grid fins were tested to evaluate the effects of span, chord and cell spacing (see Figure 10). Sample results are shown in Figure 11. The general trends for subsonic and supersonic speeds are as expected. The more dense cell spacing and the greater span produce higher fin normal force. However, at transonic speeds, span seems to be the discriminate, which most likely confirms the hypothesis that the grid fins experience a choked condition.

A point about linear aerodynamic scaling of grid fins can be illustrated with this data. If one calculates the total area of all lifting surfaces of each grid fin tested (within the set of five shown in Figure 10), and then references the normal force coefficients to the projected lifting area for each fin, then one will find that the normal force coefficient does not scale linearly with the projected lifting area. Keep in mind that the grid fins tested (Figure 10) are not a linear scale of one another, but are totally independent. Therefore, one should not expect to demonstrate a linear scaling law with this data. A set of three linearly scaled grid fins have been fabricated and are planned for future wind tunnel testing.

3.6 Grid Fin Center of Pressure

The spanwise and chordwise centers of pressure for grid fins are comparable to planar fins. Figure 12 presents center of pressure data, at zero angle of attack, for Mach numbers ranging 0.5 to 3.5. (Note that the center of pressure measurements are based on fin balance data with the grid fins in the presence of a body.) The chordwise center of pressure for grid fins is very similar to that for planar fins, generally varying between the 25% and 50% of the chord length. For the more practical grid fin design (G1), the chordwise center of pressure variation is relatively constant at approximately 30% of the chord. This result indicates a very small hinge moment for grid fin controls, which is supported by test data. The spanwise center of pressure is approximately at the 40% to 45% of span, which is essentially the same as planar fins.

3.7 Sweep Effects

One of the more curious characteristics of grid fins is the correlation of fin normal force and axial force with fin sweep angle (forward or aft). The definition of grid fin sweep angle is shown in Figure 13. A wind tunnel test was conducted to examine the effect of sweeping the grid fins forward and backward. Results from the test indicated that sweeping a fin forward or backward gives essentially the same result. A correlation of fin normal force (from fin balance data) with sweep angle is shown in Figure 14 for Mach numbers 0.5, 1.1 and 2.5. Note that fin curvature effects are small.

A correlation of fin axial force (from main balance data) is shown in Figure 15. For sweep angles of +/- 30 degrees, the axial force is amplified by a factor of 2 to 3, while the fin normal force is essentially unchanged. This result indicates that grid fins could be swept forward or backward to increase fin axial force, while maintaining fin normal force. The utility of the sweep parameter might be in using the grid fins as drag brakes while maintaining stability.

3.8 Grid Fins Versus Planar Fins

One of the most asked questions about grid fins is: "How do grid fins compare to planar fins?". A qualitative assessment of that question is shown in Figures 16 and 17. Figure 16 presents a grid fin and a planar fin that have essentially the same fin normal force at subsonic speeds and low supersonic speeds. Both fins were wind tunnel tested on an ogive cylinder body of revolution at Mach numbers 0.8, 1.8 and 2.5. The reference area for both fins is the cylinder cross sectional area. The data indicates that the grid fin produces the same normal force as the planar fin at low speeds, and approximately 50% more normal force at Mach 2.5. At the higher supersonic speeds, the grid fin normal force tends to be more linear with angle of attack than the planar fin. This characteristic has been observed in other grid fin test data at Mach numbers up to 3.5.

A qualitative assessment of axial force for two grid fins is shown in Figure 17. The frame leading and trailing edges of the grid fins are blunt. Previous test and analysis have shown that grid fin axial force can be reduced by at least 25% by simply shaping the cross section frame or reducing the web thickness. This axial force comparison is somewhat misleading, however. In general, grid fin axial force will be higher than the comparative planar fin axial force at subsonic speeds, whereas at high supersonic speeds, the comparative grid fin will be smaller, producing less axial force for the comparative normal force. Consequently, the supersonic axial force of a grid fin might be very comparable to the axial force of a planar fin which generates an equivalent amount of normal force. Further test and analysis is required to better answer this question.

4. FLIGHT TEST PROGRAM

Grid fins were first flown in the United States on rocket flight and air drop flight tests conducted in 1996. These flight tests were a culmination of an 18-month technical demonstration program involving alternate submunition dispense technology. This program, known as Bat-On-A-Rocket (BOAR), was conducted by the Army Aviation and Missile Command, Missile Research Development and Engineering Center during 1995 and 1996. Photographs of the rocket and air drop flight test hardware are presented in Figures 18 and 19, respectively.

Figure 20 presents an illustration of the rocket flight test scenario. The forward section (warhead) of the rocket was equipped with grid fins which were stowed by folding forward into a cavity within the cylindrical boundary of the rocket body. The warhead section was attached to a tactical rocket booster and launched from a tactical launcher with cylindrical launch tubes. During the flight, the warhead was separated from the booster (near apogee), the grid fins were deployed, and the warhead section continued on the trajectory. Curved grid fins were used in this application because of their excellent packaging characteristics. Grid fins were folded forward into cavities and shielded with cover plates. Grid fin deployment occurred after booster separation in order to stabilize and despin the warhead.

The center of pressure of the warhead section with grid fins was measured in wind tunnel tests of the model shown in Figure 21. Empirical prediction methods were used to design the grid fins and to predict flight characteristics. Comparisons between original design predictions and measured center of pressures are shown in Figure 22. These results indicate that the center of pressure of a rocket, or missile, with grid fins can be accurately estimated with engineering level aerodynamic prediction methods.

In the rocket flight test, the rocket was forced to spin in flight (about 6-8 rps) to minimize flight perturbations. After stage separation, the warhead section was required to de-spin immediately. The de-spin characteristics (roll damping) of grid fins were unknown at that time, since no grid fin roll damping data were available. Consequently, a roll damping wind tunnel test was conducted to measure the roll damping characteristics of the grid fins which were specifically designed for the rocket flight test.

Figure 23 presents a comparison between original predicted and measured roll damping coefficient data. The prediction data is based on classical aerodynamic prediction methods. Although refinements using test results would lead to improvements in prediction method accuracy, the results indicate that grid fins have very effective roll damping characteristics and that the prediction of grid fin roll damping is clearly within the realm of classical aerodynamic analysis methods.

As mentioned earlier, an airdrop test of a simulated rocket warhead section was also conducted. Release conditions corresponded to Mach 0.9 at an altitude of 40,000 feet. The flight hardware shown in Figure 19 flew a ballistic trajectory and successfully dispensed two submunitions at designated altitudes.

5. CONCLUSIONS

Based on a series of nine wind tunnel tests of grid fins, two flight tests of rockets stabilized with grid fins, and one air drop flight test stabilized with grid fins, certain conclusions can be stated. It is clear that grid fins offer some definite advantage over conventional planar fins. These advantages include:

- Excellent supersonic control characteristics.
- Compact storage, relatively easy to deploy.
- Excellent high angle of attack characteristics.
- Small center of pressure variations over wide Mach number ranges.
- Ability to tailor drag using frame shaping.
- Ability to alter lift and drag using fin sweep.
- High strength-to-weight ratios.
- Small hinge moments minimize actuator requirements.

The aerodynamic characteristics of grid fins, coupled with their excellent storage characteristics, make them particularly attractive devices for canister launched missiles, ship launched missiles, multi-stage missiles, artillery launched munitions, missiles designed for deployment from internal weapon bays, compressed carriage weapons and dispensed submunitions. Some of the more notable unknowns concerning grid fins include: minimum drag levels, radar cross section and mass production manufacturing techniques.

Table 1. Grid Fin Wind Tunnel Test Summary

Test Date	Test Facility	Test Objective
AUG 1985	Vought HSWT (Grand Prairie, TX)	Exploratory investigation of flat grid fins
FEB 1989	Vought HSWT (Grand Prairie, TX)	Exploratory investigation of curved grid fins and sweep effects
OCT 1993	NTS 4 × 4 TWT (Saugus, CA)	Investigation of grid fin design as control surface for TACAWS
NOV 1993	NTS 4 × 4 TWT (Saugus, CA)	Investigation of grid fin drag reduction techniques
SEP 1994	NTS 4 × 4 TWT (Saugus, CA)	Investigation of new grid fin designs (parametric variation)
OCT 1995	NTS 4 × 4 TWT (Saugus, CA)	BOAR warhead vehicle application BOAR airdrop vehicle application
OCT 1995	Loral Vought Systems HSWT (Grand Prairie, TX)	Roll damping investigation of BOAR and other grid fin designs
NOV 1996	USAF SARL (WPAFB, OH)	Subsonic splitter plate test for 2-D aerodynamics of grid fin designs
MAY 1997	USAF SARL (WPAFB, OH)	Subsonic investigation of grid fin aerodynamics in presence of body

Table 2. Grid Fin Configuration Summary

Configuration Code	Fin Description	Wind Tunnel Test
G1 G2	Flat Grid Fin - dense webbing Flat Grid Fin - sparse webbing	Vought 1985
G3 G4 G5	Fin G1 with revised attachment structure Fin G1 with convex curvature Fin G1 with concave curvature	Vought 1989
G6 G7 G8 G9 G10 G11	Baseline fin with no frame shaping Baseline with modified double wedge frame Baseline with modified single wedge frame Baseline with half diamond frame Fin G7 with thicker web Baseline with thin modified double wedge frame	NTS 1993 USAF SARL 1996 USAF SARL 1997
G12 G13 G14 G15 G16	Fin G6 with modifications Fin G12 with longer span Fin G12 with denser webbing Fin G13 with sparser webbing Fin G13 with increased web depth	NTS 1994 NTS 1995 Loral Vought 1995 USAF SARL 1996 USAF SARL 1997
G17 G18	MLRS BOAR flight hardware - 50.35% scale Fin G17 with longer span	NTS 1995 Loral Vought 1995 USAF SARL 1996 USAF SARL 1997
G19	MLRS BOAR flight hardware - 24.77% scale	Not Tested
G20	TACAWS Grid Fin Design	NTS 1993 USAF SARL 1996 USAF SARL 1997
G21 G22	Non-orthogonal design with dense webbing Non-orthogonal design with less dense webbing	USAF SARL 1996 USAF SARL 1997
G23 G24	Research fin, curved with long span Research fin, curved with short span	USAF SARL 1996
G25 G26	Fin G6 with attach structure on side of frame Fin G13 with attach structure on side of frame	USAF SARL 1996

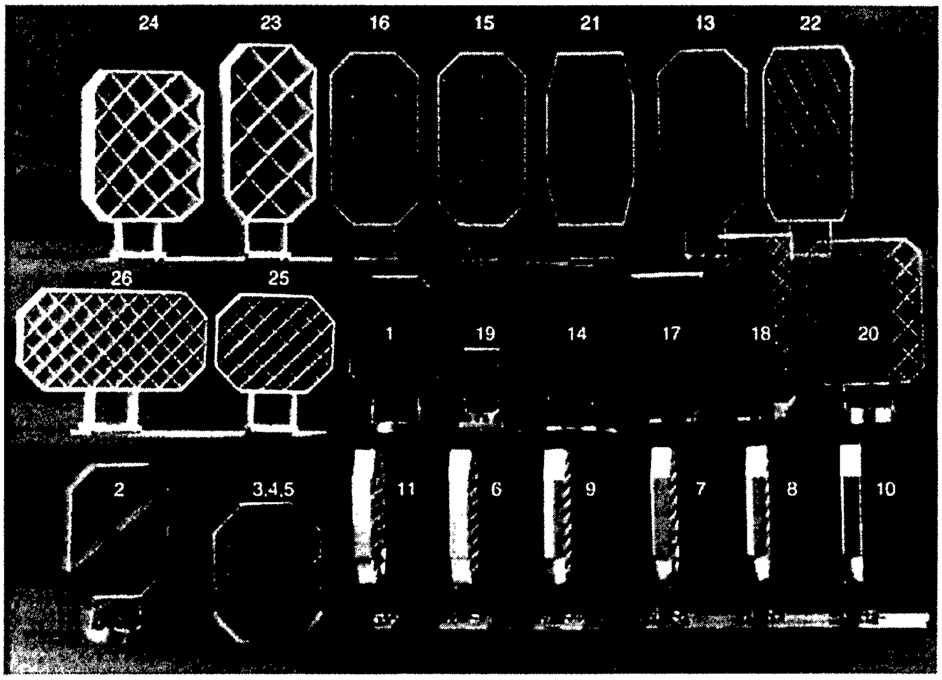


Figure 1. Photograph of Grid Fin Designs

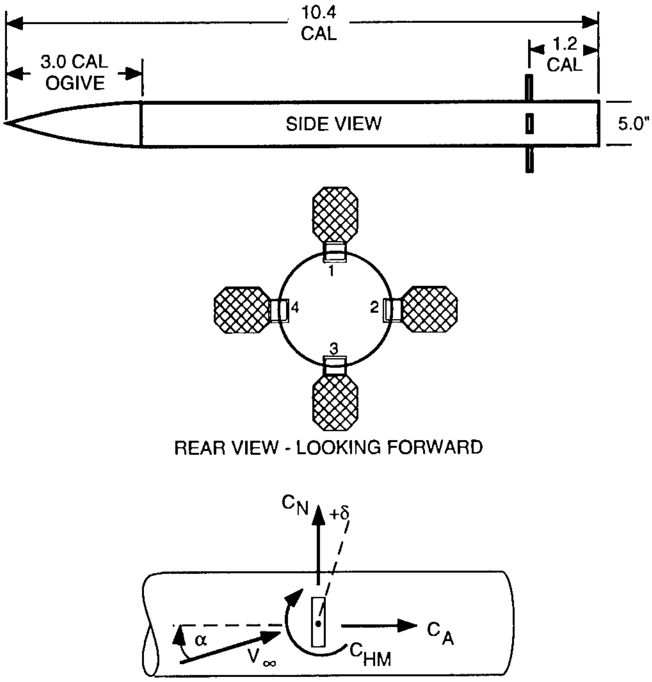


Figure 2. Model and Sign Conventions

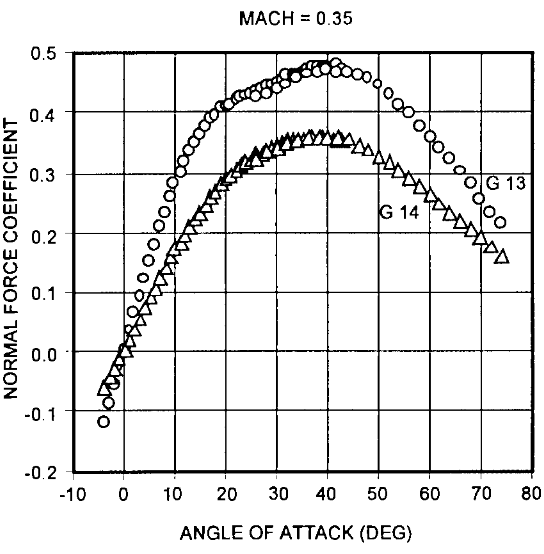


Figure 3. High Angle of Attack Characteristics

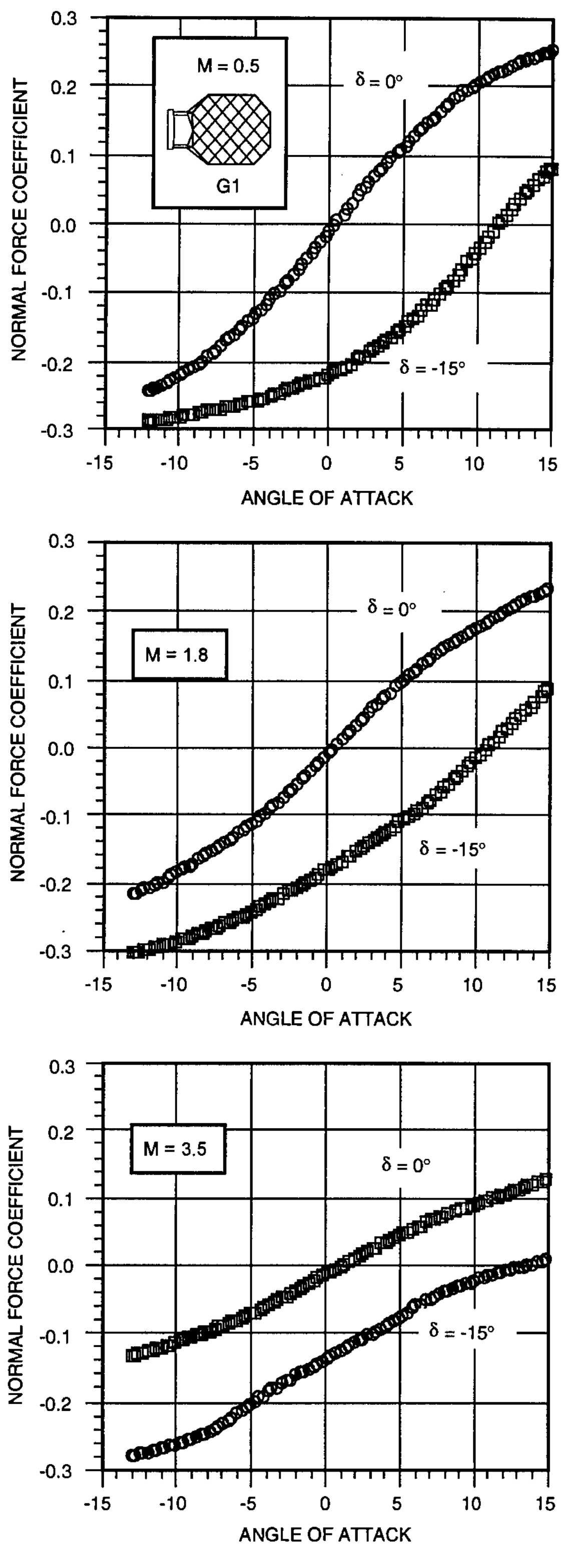


Figure 4. Normal Force Control Characteristics

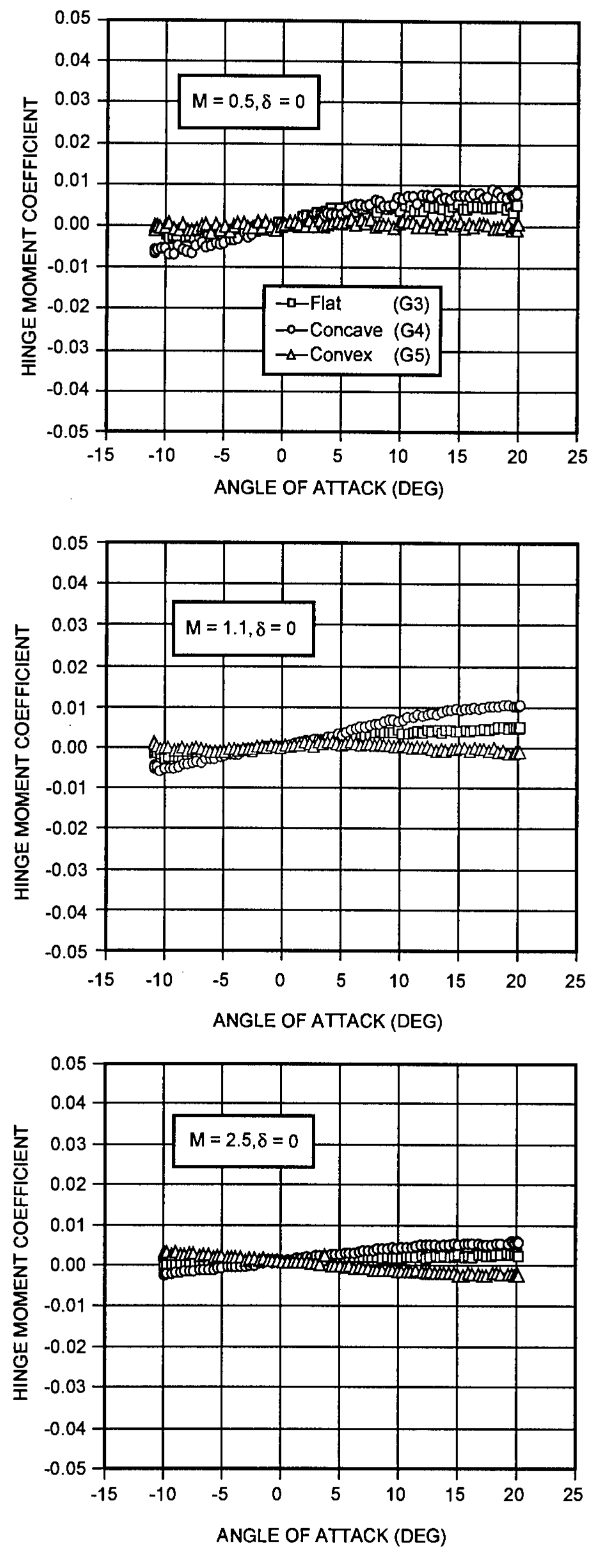
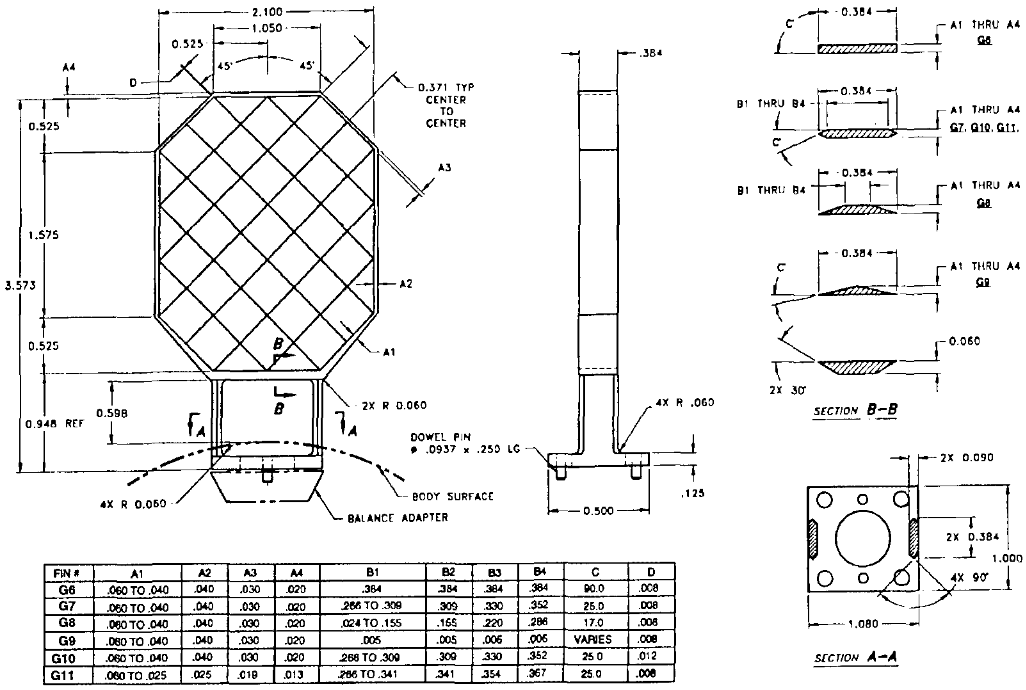


Figure 5. Curvature Effects on Hinge Moments



FIN #	A1	A2	A3	A4	B1	B2	B3	B4	C	D
G6	.060 TO .040	.040	.030	.020	.384	.384	.384	.384	90.0	.008
G7	.060 TO .040	.040	.030	.020	.266 TO .309	.309	.330	.352	25.0	.008
G8	.060 TO .040	.040	.030	.020	.024 TO .155	.155	.220	.286	17.0	.008
G9	.060 TO .040	.040	.030	.020	.005	.005	.006	.006	VARIES	.008
G10	.060 TO .040	.040	.030	.020	.268 TO .309	.309	.330	.352	25.0	.012
G11	.060 TO .025	.025	.019	.013	.266 TO .341	.341	.354	.367	25.0	.008

Figure 6. Grid Fin Designs Used for Drag Evaluation Test

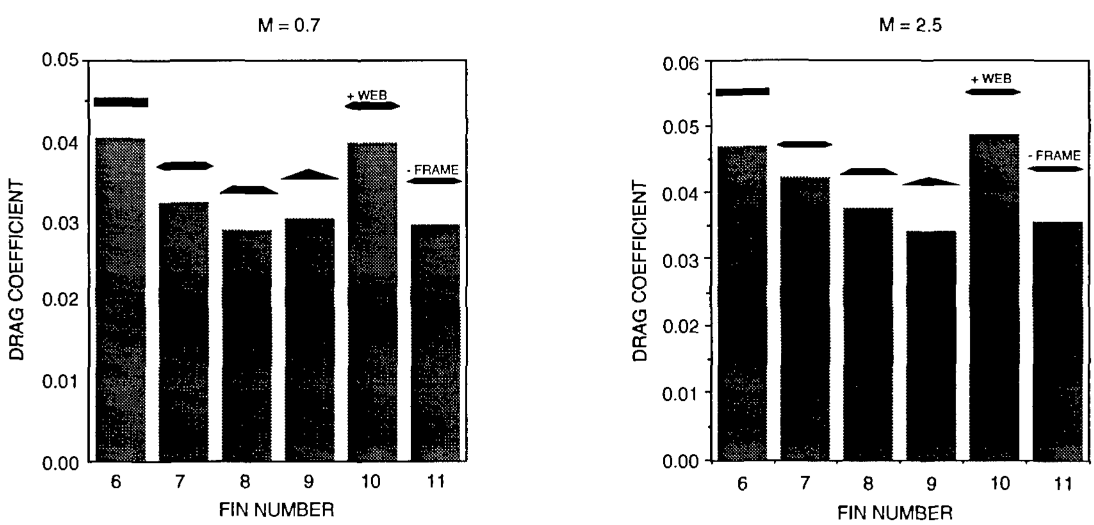


Figure 7. Drag Evaluation Test Results

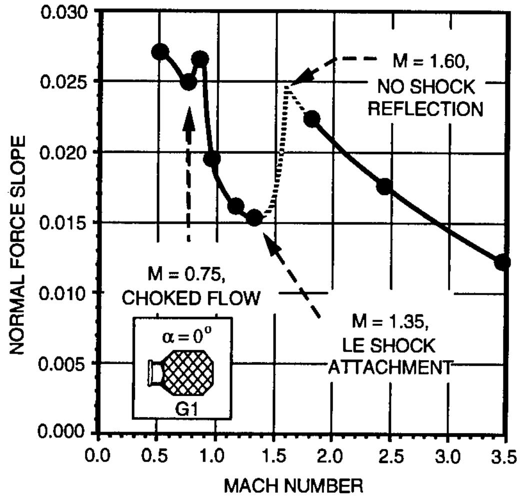


Figure 8. Normal Force Slope Variation

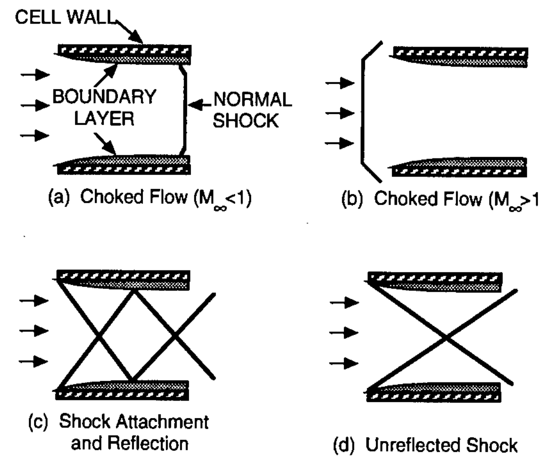


Figure 9. Grid Fin Flow Regimes

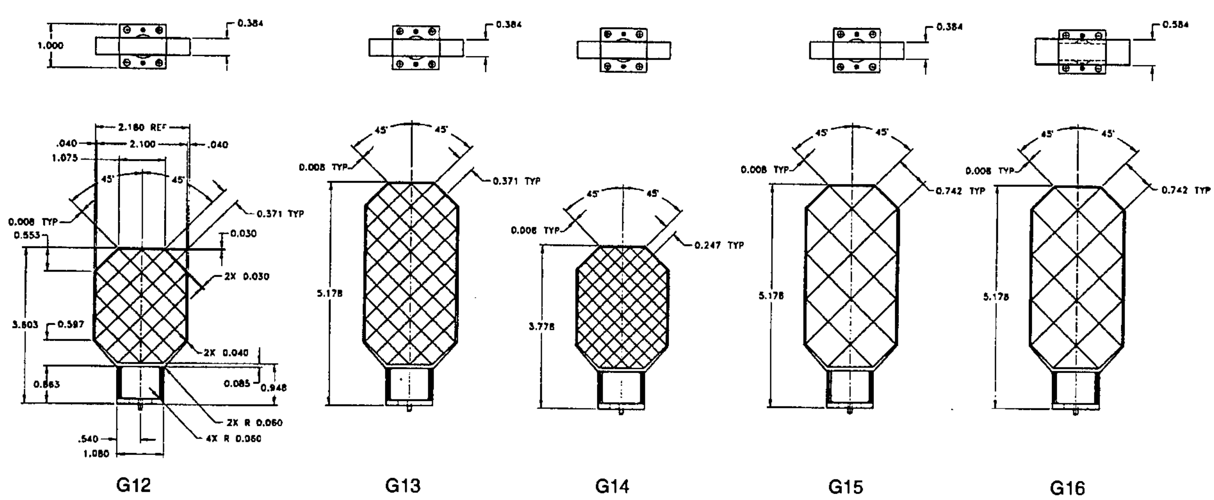


Figure 10. Grid Fins Used for Parametric Design Evaluation Testing

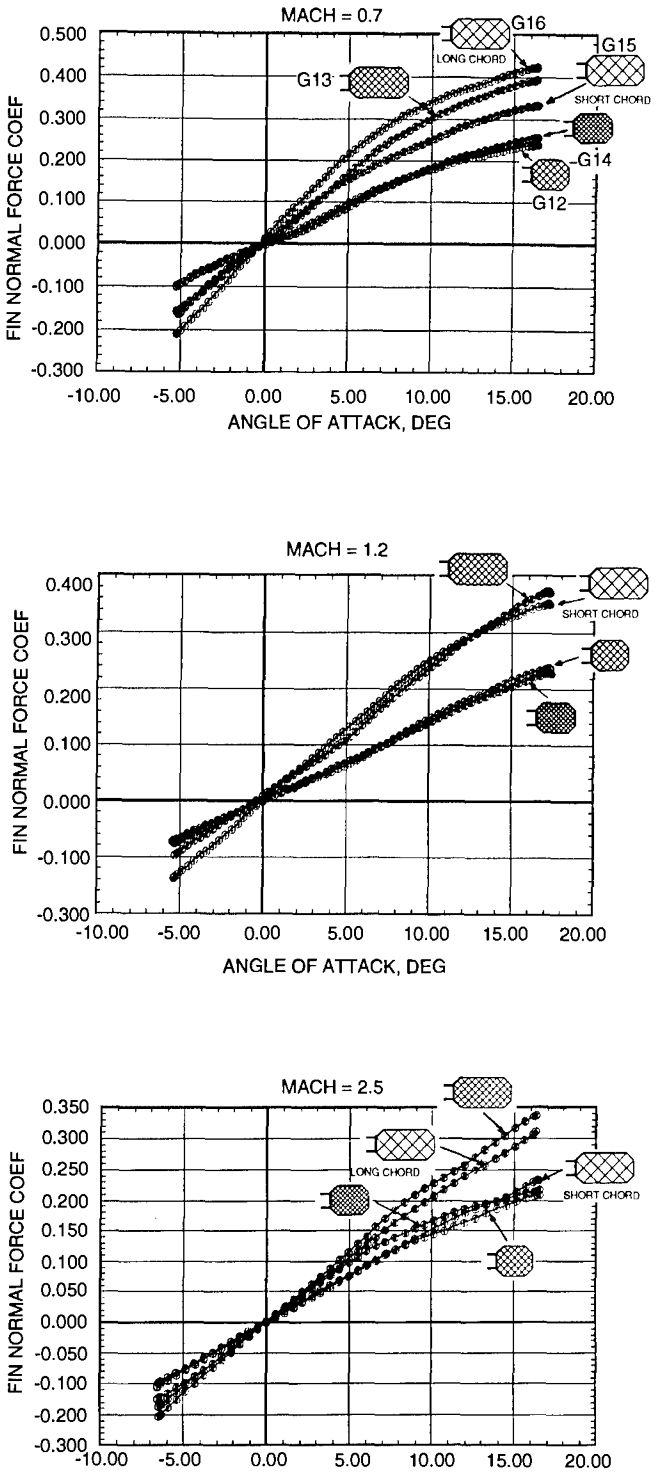


Figure 11. Parametric Design Test Results

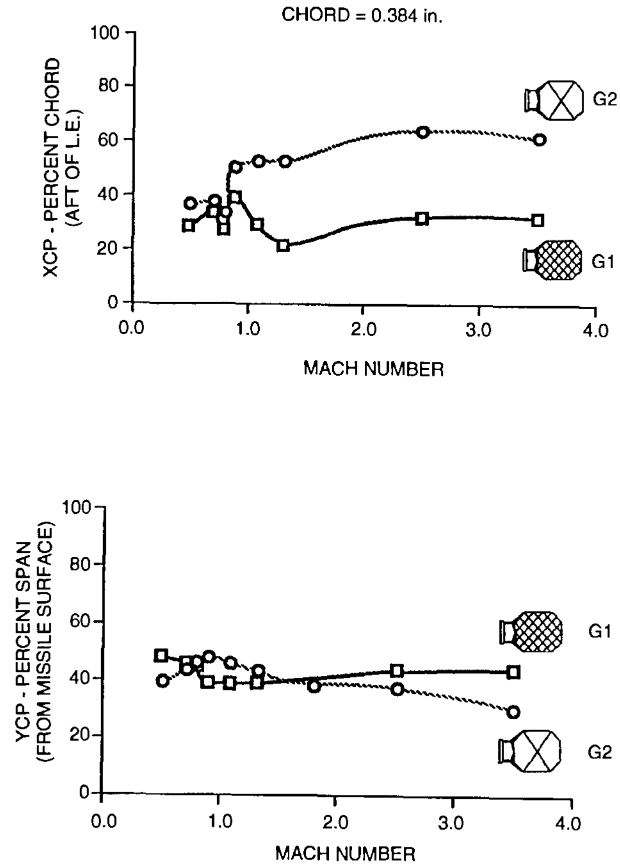


Figure 12. Grid Fin Center of Pressure Variations

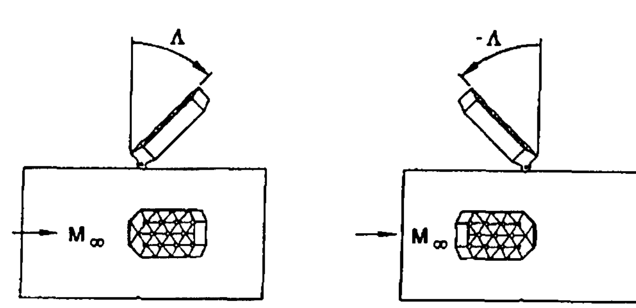


Figure 13. Sweep Sign Conventions

Figure 14. Sweep Effects on Normal Force

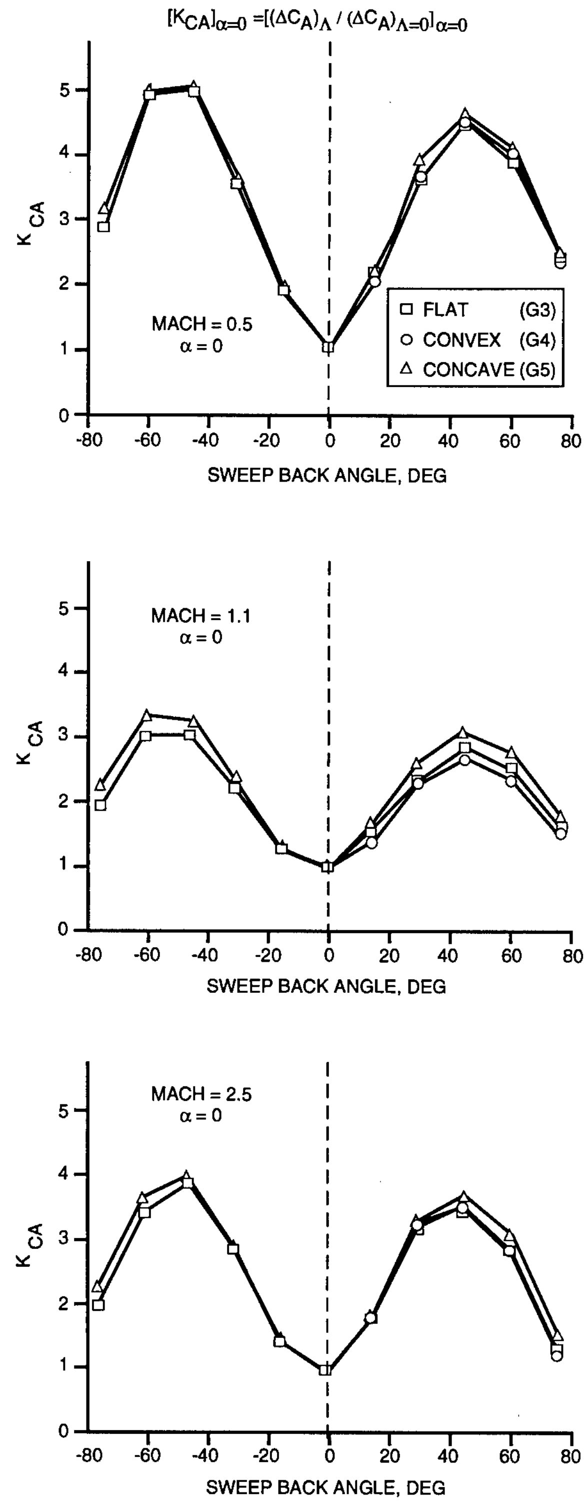


Figure 15. Sweep Effects on Axial Force

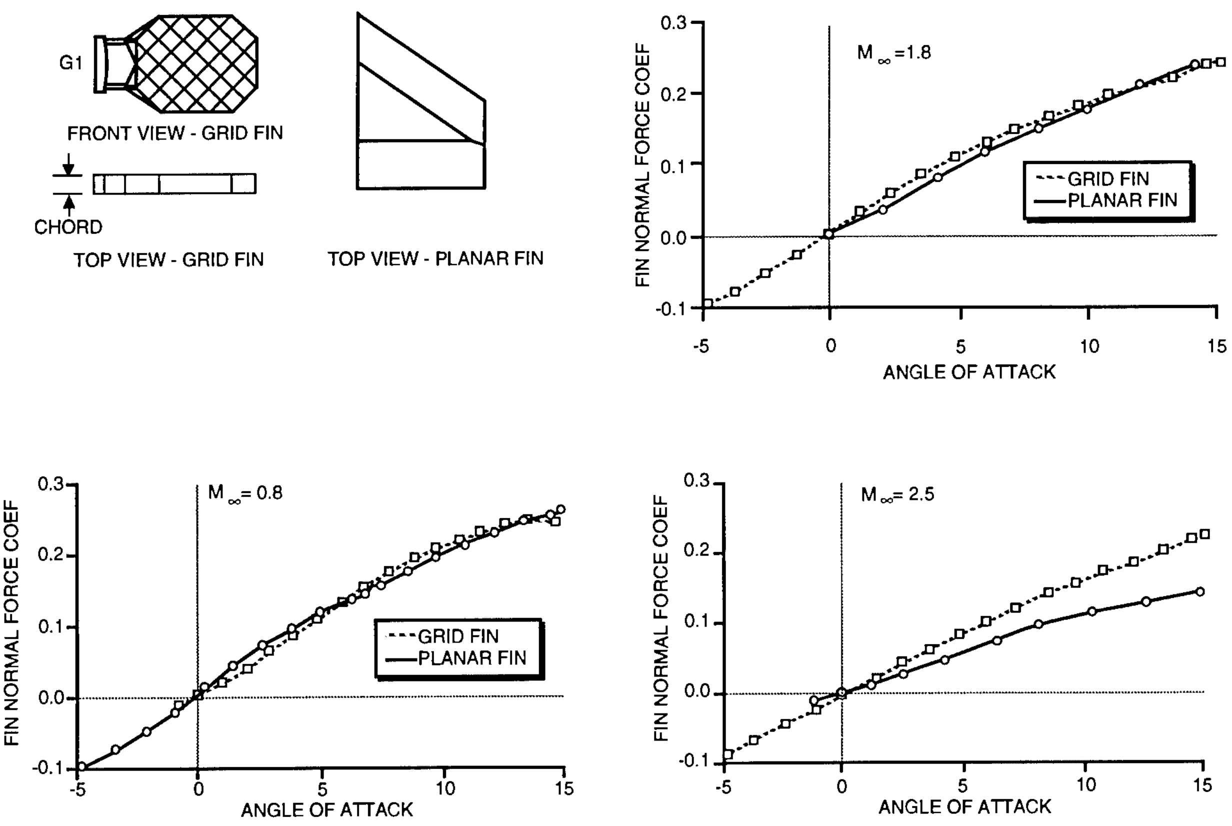


Figure 16. Grid Fin / Planar Fin Normal Force Comparisons

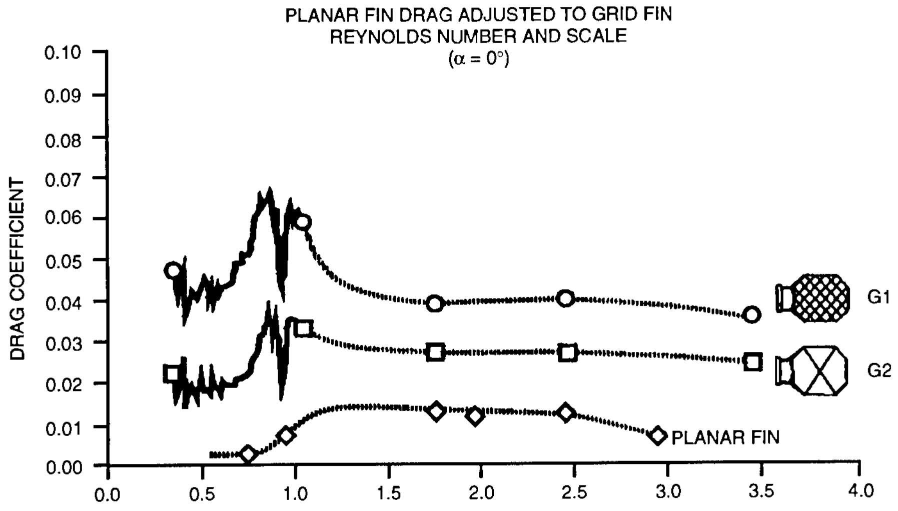


Figure 17. Grid Fin / Planar Fin Axial Force Comparison

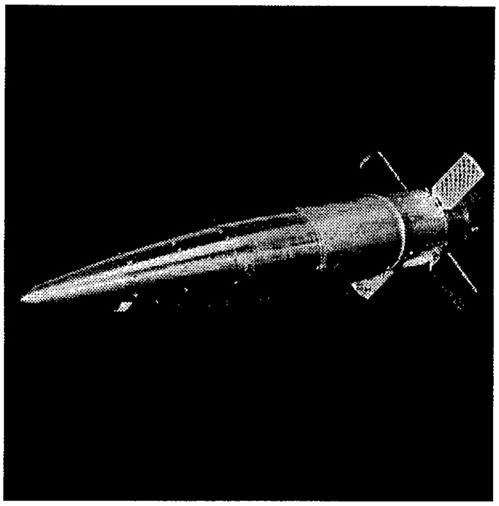


Figure 18. BOAR Rocket Warhead Flight Hardware

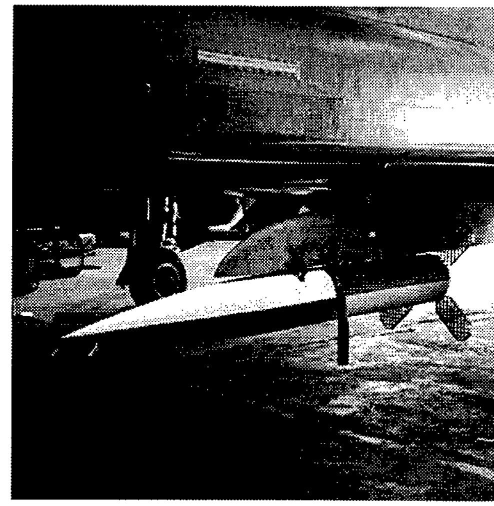


Figure 19. BOAR Airdrop Flight Hardware

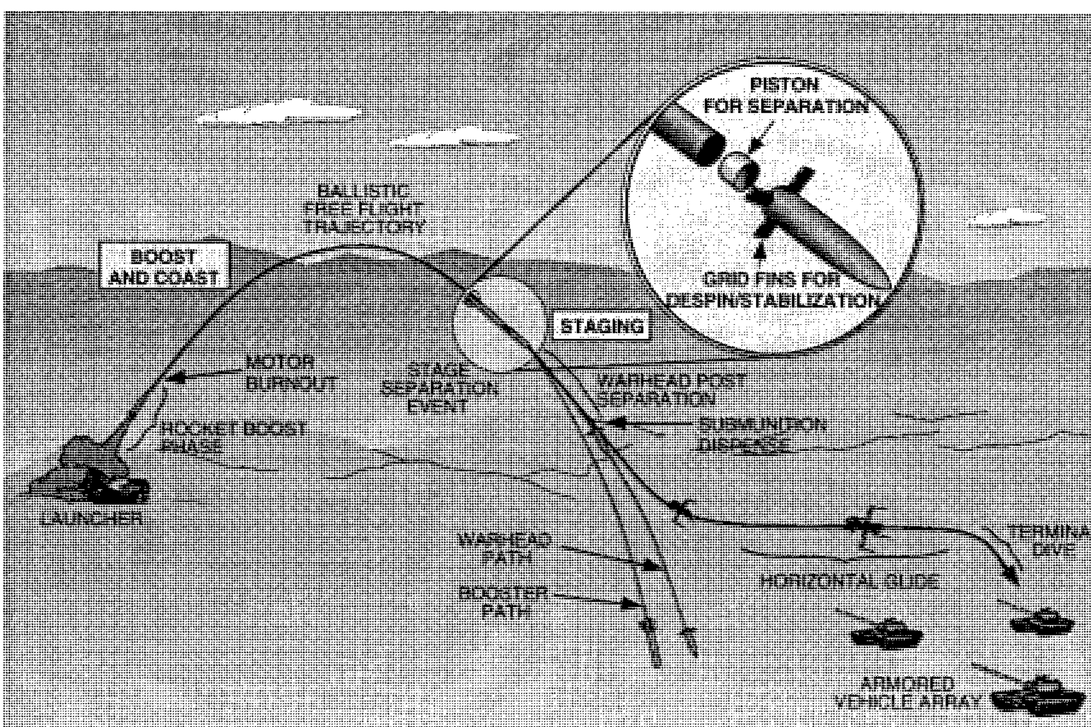


Figure 20. Bat-On-A-Rocket (BOAR) Flight Scenario

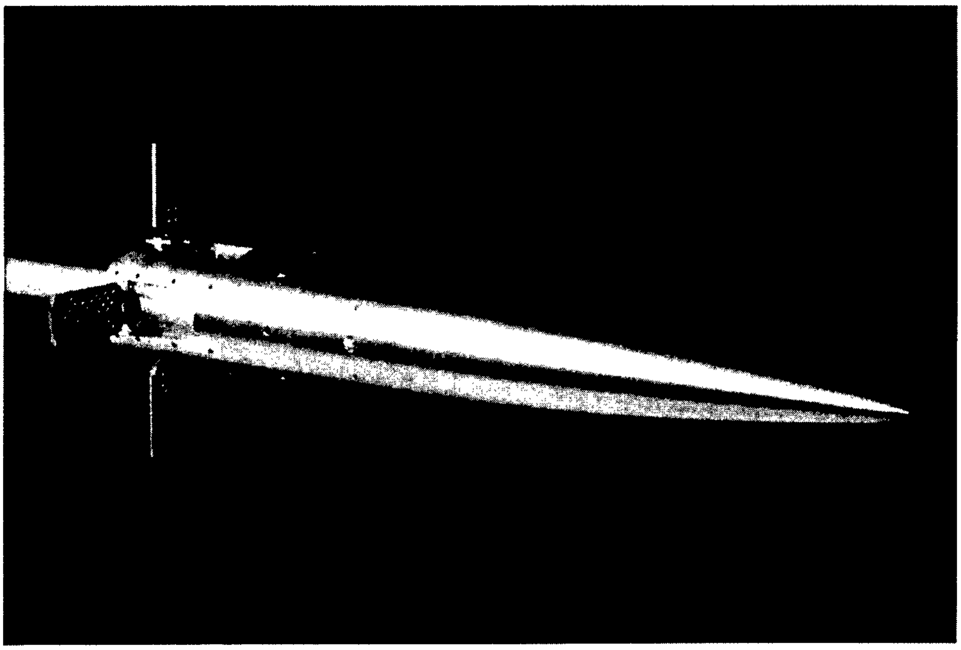


Figure 21. BOAR Warhead Section Wind Tunnel Model

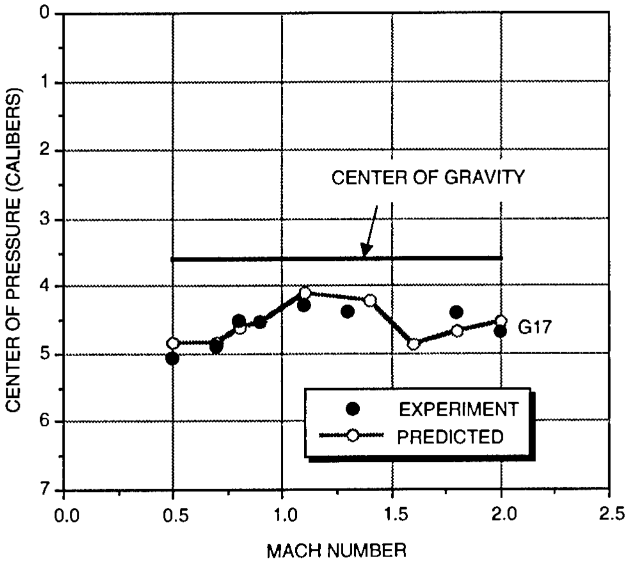


Figure 22. Warhead Center of Pressure Comparisons

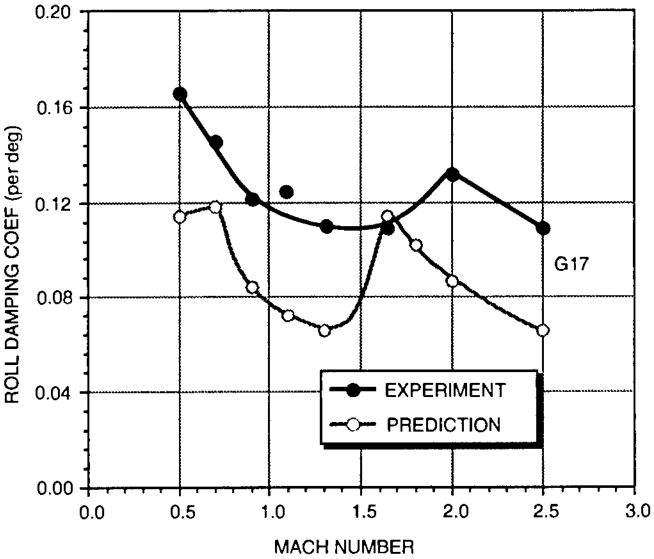


Figure 23. Warhead Roll Damping Comparisons